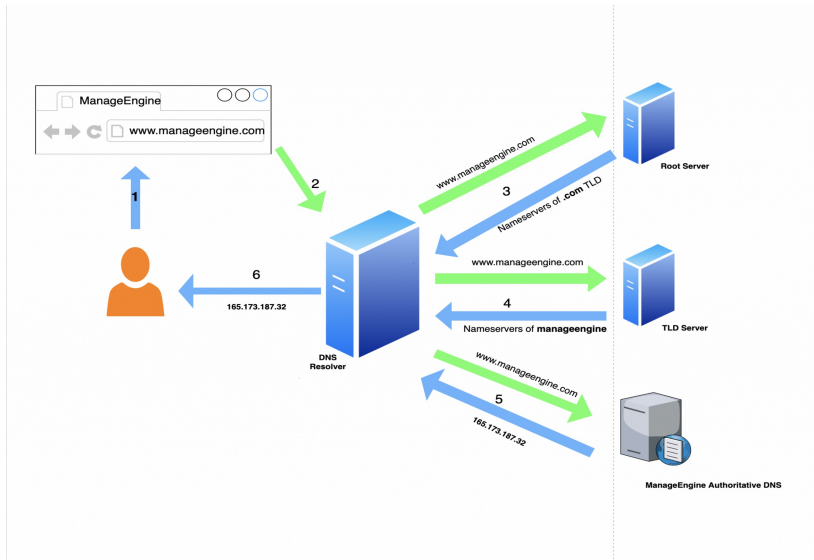


What is DNS?

DNS (Domain Name System) is like the internet's phonebook. It translates human-friendly domain names (e.g., google.com) into IP addresses (e.g., 142.250.190.78) that computers use to locate and communicate with each other.

How Does DNS Work?

When you enter a website URL in your browser, the DNS process follows these steps:



1. User Request

You type `example.com` into your web browser.

2. DNS Resolver (Recursive Resolver)

Your request is sent to a DNS resolver, usually provided by your ISP (Internet Service Provider) or services like Google DNS (8.8.8.8) or Cloudflare DNS (1.1.1.1).

3. Checking the Cache

If the resolver has the IP address stored in its cache, it returns the result immediately. If not, it queries other DNS servers.

4. Querying the DNS Hierarchy

If the resolver doesn't find the IP in its cache, it follows these steps:

Root DNS Server: Directs the request to the relevant Top-Level Domain (TLD) server (e.g., `.com`, `.org`).

TLD DNS Server: Points to the domain's authoritative name server.

Authoritative DNS Server: Provides the actual IP address for `example.com`.

5. Returning the IP Address

The resolver gets the IP address and sends it back to the user's browser.

6. Website Loads

The browser connects to the web server at that IP address, retrieves the webpage, and displays it.

Types of DNS Servers:

DNS Resolver: Finds the IP by querying other DNS servers.

Root DNS Server: Directs queries to TLD servers.

TLD DNS Server: Manages domains under a specific TLD (`.com`, `.net`).

Authoritative DNS Server: Stores the actual domain-to-IP mapping.

Types of DNS Records

A Record: Maps domain names to IPv4 addresses.

AAAA Record: Maps domain names to IPv6 addresses.

CNAME Record: Aliases one domain to another.

MX Record: Specifies mail servers for email routing.

TXT Record: Stores text information for verification/security.